

Mass/Spring/Damper Systems Part 1

EGR 345

GVSU

Outline

- review
- mass/spring/damper systems
 - initial condition response
 - Laplace with a second order denominator
 - Laplace with complex roots
 - critical damping
 - defining ζ (zeta)

Review: Two key characteristics of a dynamic system

- one “practical” and one mathematical
- vibrations are a transient response that can take a while to die out

↑
another practical indicator

→ transient response matters
→ input/output relationship depends on diff. eq.